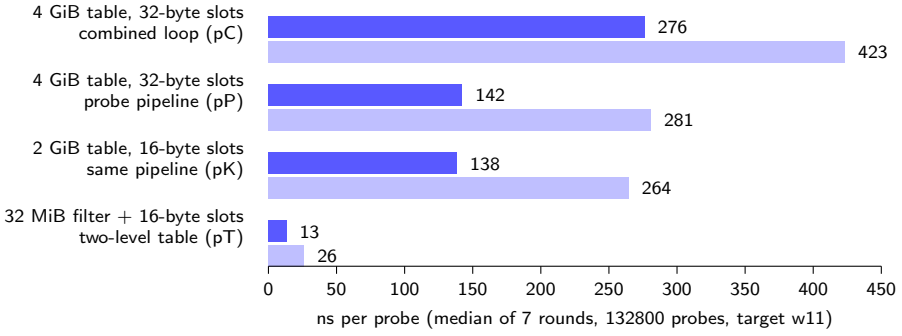
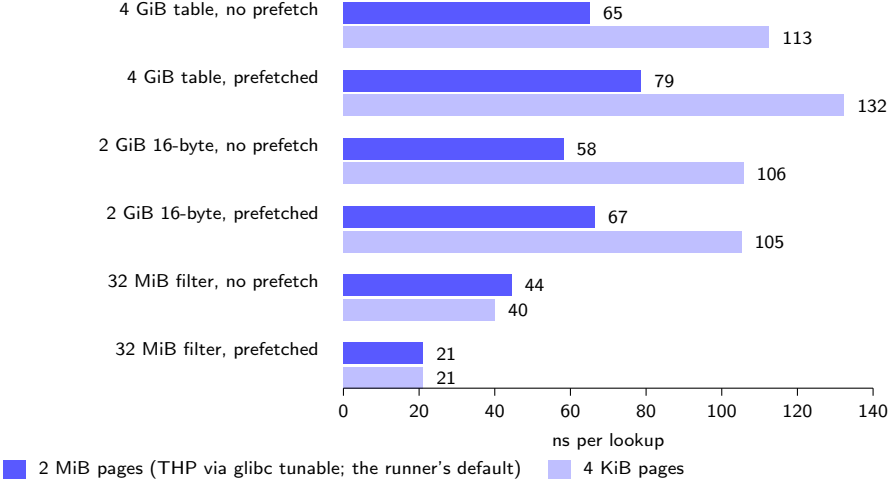
In-situ lookup cost, marginal ns per probe (over keys alone, 203 ns)



Lookup-only loops over the same keys (batches of one state's 332 keys)



Diagnostic: cloud-agent VM (shared Intel Xeon guest), profiler variants of the derived producer. Not a speedup claim.